

Media Player Application

KOTLIN

```
package com.example.media

import android.os.Bundle
import android.media.MediaPlayer
import android.widget.Button
import android.widget.SeekBar
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat

class MainActivity : AppCompatActivity() {

    private lateinit var mediaPlayer: MediaPlayer
    private lateinit var seekBar: SeekBar

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val playButton = findViewById<Button>(R.id.buttonPlay)
        val pauseButton = findViewById<Button>(R.id.buttonPause)
        val forwardButton = findViewById<Button>(R.id.buttonForward)
        val backwardButton = findViewById<Button>(R.id.buttonBackward)
        seekBar = findViewById(R.id.seekBar)

        // Audio file placed in res/raw folder (example: song.mp3)
        mediaPlayer = MediaPlayer.create(this, R.raw.song)

        seekBar.max = mediaPlayer.duration

        playButton.setOnClickListener {
            mediaPlayer.start()
        }

        pauseButton.setOnClickListener {
            mediaPlayer.pause()
        }

        forwardButton.setOnClickListener {
            val currentPosition = mediaPlayer.currentPosition
            mediaPlayer.seekTo(currentPosition + 5000) // forward 5 sec
        }
    }
}
```

```

backwardButton.setOnClickListener {
    val currentPosition = mediaPlayer.currentPosition
    mediaPlayer.seekTo(currentPosition - 5000) // backward 5 sec
}

// Update SeekBar continuously
Thread {
    while (mediaPlayer != null) {
        try {
            if (mediaPlayer.isPlaying) {
                seekBar.progress = mediaPlayer.currentPosition
            }
            Thread.sleep(1000)
        } catch (e: Exception) {
            e.printStackTrace()
        }
    }
}.start()

seekBar.setOnSeekBarChangeListener(object : SeekBar.OnSeekBarChangeListener {
    override fun onProgressChanged(seekBar: SeekBar?, progress: Int, fromUser: Boolean) {
        if (fromUser) {
            mediaPlayer.seekTo(progress)
        }
    }

    override fun onStartTrackingTouch(seekBar: SeekBar?) {}

    override fun onStopTrackingTouch(seekBar: SeekBar?) {}
})

override fun onDestroy() {
    super.onDestroy()
    mediaPlayer.release()
}
}

```

XML

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:layout_height="match_parent"
    android:padding="16dp">

```

```
<Button
    android:id="@+id/buttonPlay"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Play" />
```

```
<Button
    android:id="@+id/buttonPause"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Pause" />
```

```
<Button
    android:id="@+id/buttonForward"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Forward" />
```

```
<Button
    android:id="@+id/buttonBackward"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Backward" />
```

```
<SeekBar
    android:id="@+id/seekBar"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />
```

```
</LinearLayout>
```